KAIZEN MATHS

Lesson starter

Hyperbolic Functions

Inverse Hyperbolic Functions

Name

S1

Write each inverse hyperbolic value in logarithmic form.

Write arsinh 3 in logarithmic form.

S2

Differentiate each inverse hyperbolic function.

Differentiate y = artanh(3x).

S3

Solve each inverse hyperbolic equation exactly.

Solve artanh x = ln 5.

S4

Write each inverse hyperbolic value in logarithmic form.

Write arsinh 2 in logarithmic form.

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ALG-6FDD-L4 · S

Show your working

KAIZEN MATHS

Exit ticket

Hyperbolic Functions

Inverse Hyperbolic Functions

Name _____

E1

Differentiate each inverse hyperbolic function.

Differentiate $y = \text{arcosh}(2x)$.

E2

Solve each inverse hyperbolic equation exactly.

Solve $\text{arsinh } x = \ln 5$.

E3

Write each inverse hyperbolic value in logarithmic form.

Write $\text{arsinh } 4$ in logarithmic form.

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ALG-6FDD-L4 · E

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ALG-6FDD-L4 · E

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Teacher answers & working

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S1

Write each inverse hyperbolic value in logarithmic form.

Write arsinh 3 in logarithmic form.

Answer:

$\ln(3 + \sqrt{10})$

Working:

Use arsinh x = ln(x + √(x² + 1)).

Substitute x = 3.

arsinh 3 = ln(3 + √(9 + 1)) = ln(3 + √10).

S2

Differentiate each inverse hyperbolic function.

Differentiate y = artanh(3x).

Answer:

$\frac{dy}{dx} = \frac{3}{1-9x^2}$

Working:

Use $\frac{d}{dx}$artanh u = $\frac{u'}{1-u^2}$.

Here u = 3x, so u' = 3.

$\frac{dy}{dx} = \frac{3}{1-3x^2} = \frac{3}{1-9x^2}$.

S3

Solve each inverse hyperbolic equation exactly.

Solve artanh x = ln 5.

Answer:

$x = \frac{12}{13}$

Working:

Apply tanh to both sides.

x = tanh(ln 5).

Use $e^{\ln 5} = 5$ and $e^{-\ln 5} = \frac{1}{5}$.

$x = \frac{5-\frac{1}{5}}{5+\frac{1}{5}} = \frac{12}{13}$.

Teacher answers & working

Hyperbolic Functions · Inverse Hyperbolic Functions

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S4

Write each inverse hyperbolic value in logarithmic form.

Write arsinh 2 in logarithmic form.

Answer:

$\ln(2 + \sqrt{5})$

Working:

Use arsinh x = ln(x + √(x² + 1)).

Substitute x = 2.

arsinh 2 = ln(2 + √(4 + 1)) = ln(2 + √5).

E1

Differentiate each inverse hyperbolic function.

Differentiate y = arcosh(2x).

Answer:

$\frac{dy}{dx} = \frac{2}{\sqrt{(2x-1)}\sqrt{(2x+1)}}$

Working:

Use $\frac{d}{dx}$arcosh u = $\frac{u'}{\sqrt{(u-1)}\sqrt{(u+1)}}$.

Here u = 2x, so u' = 2.

$\frac{dy}{dx} = \frac{2}{\sqrt{(2x-1)}\sqrt{(2x+1)}}$.

E2

Solve each inverse hyperbolic equation exactly.

Solve arsinh x = ln 5.

Answer:

$x = \frac{12}{5}$

Working:

Apply sinh to both sides.

x = sinh(ln 5).

Use $e^{\ln 5} = 5$ and $e^{-\ln 5} = \frac{1}{5}$.

$x = \frac{5-\frac{1}{5}}{2} = \frac{12}{5}$.

Teacher answers & working

Hyperbolic Functions · Inverse Hyperbolic Functions

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E3

Write each inverse hyperbolic value in logarithmic form.

Write arsinh 4 in logarithmic form.

Answer:

$\ln(4 + \sqrt{17})$

Working:

Use $\operatorname{arsinh} x = \ln(x + \sqrt{(x^2 + 1)})$.

Substitute $x = 4$.

$\operatorname{arsinh} 4 = \ln(4 + \sqrt{(16 + 1)}) = \ln(4 + \sqrt{17})$.